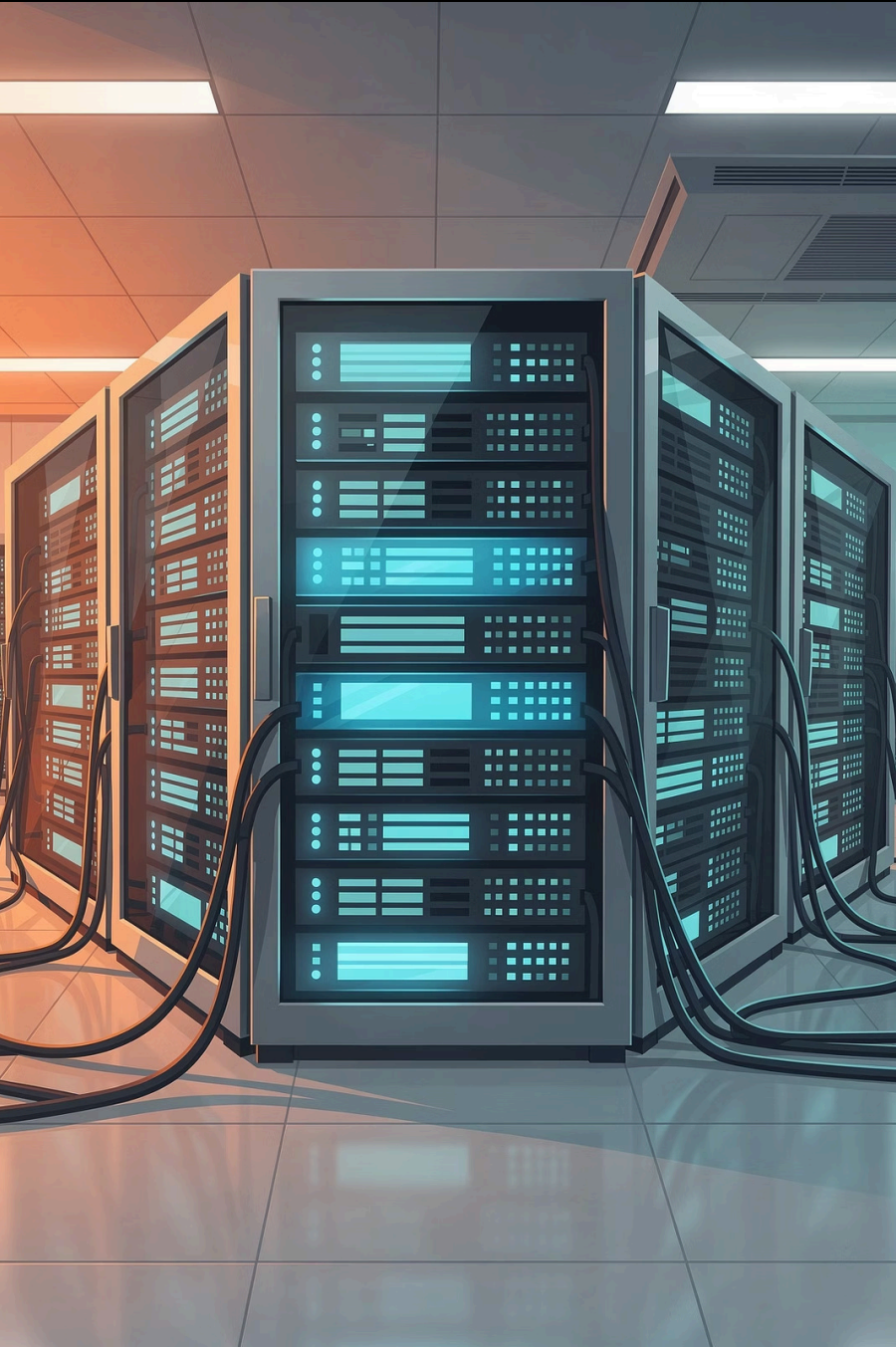




Fundamentals of Virtualization

Virtualization is one of the most important fundamental concepts of modern IT. Understanding it unlocks servers, cloud, containers, networks, and IT security. The core: **Software is used to turn one physical computer into multiple virtual computers** that behave like independent systems.

IT FUNDAMENTALS

VIRTUALIZATION

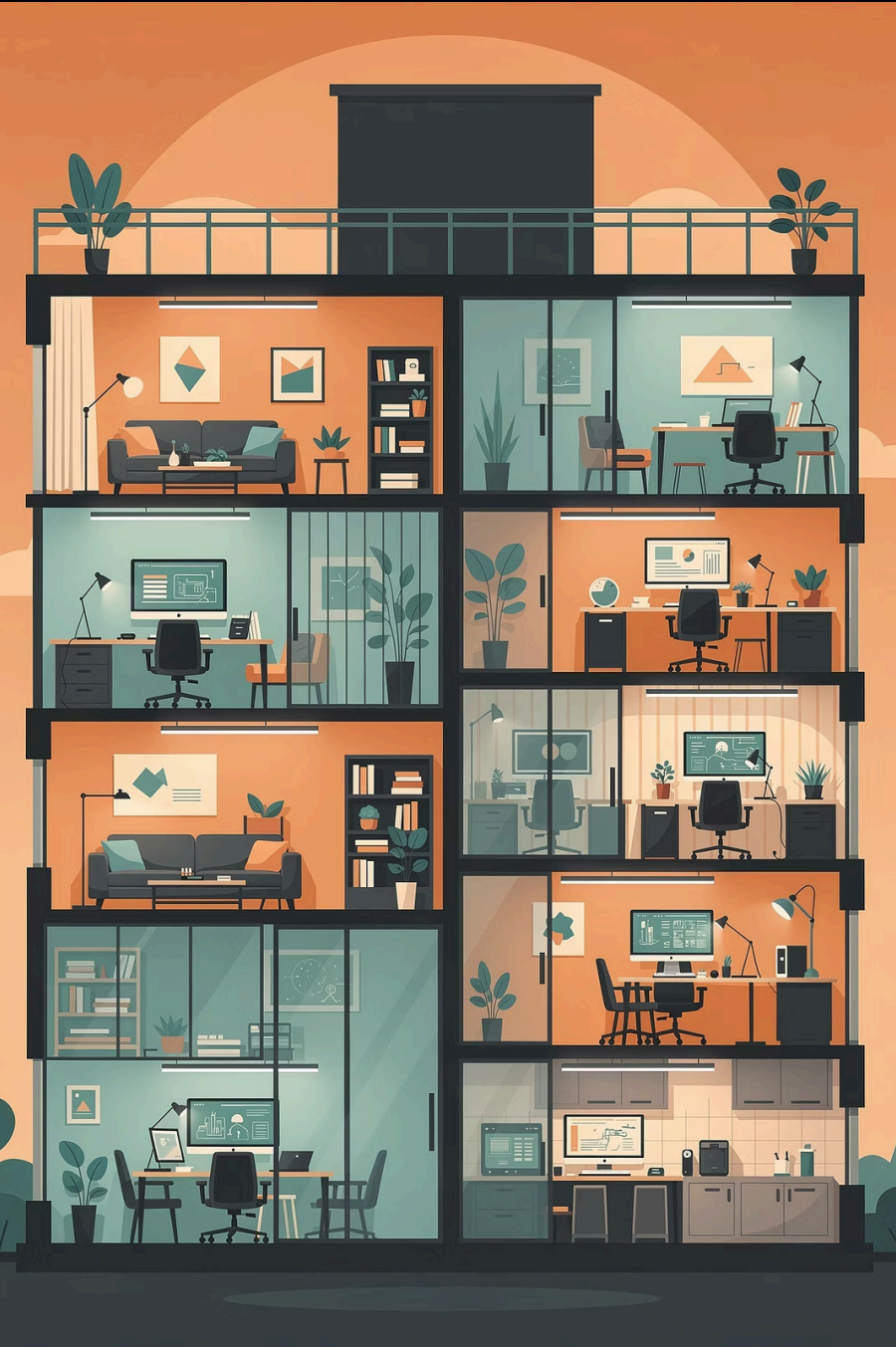
What is Virtualization?

The Core Idea

Virtualization separates the operating system's view of hardware from the actual, physical hardware. The operating system "believes" it has its own computer — in reality, it operates within a virtual environment.

The Apartment Building Model

The **physical hardware** is the building. The **virtual machines** are the apartments within it. Each apartment appears as its own space, even though all share the same building — CPU, RAM, hard drive, and network are shared without it being apparent.



Definition and Significance

Virtualization means: hardware resources are no longer used exclusively by one operating system. A software layer—the **hypervisor**—distributes these resources and makes them available to multiple virtual systems. This allows Windows, Linux, and other systems to run simultaneously on a single physical computer.

Efficiency

Servers are often only 10–20% utilized—virtualization consolidates multiple systems and makes much better use of hardware.

Flexibility

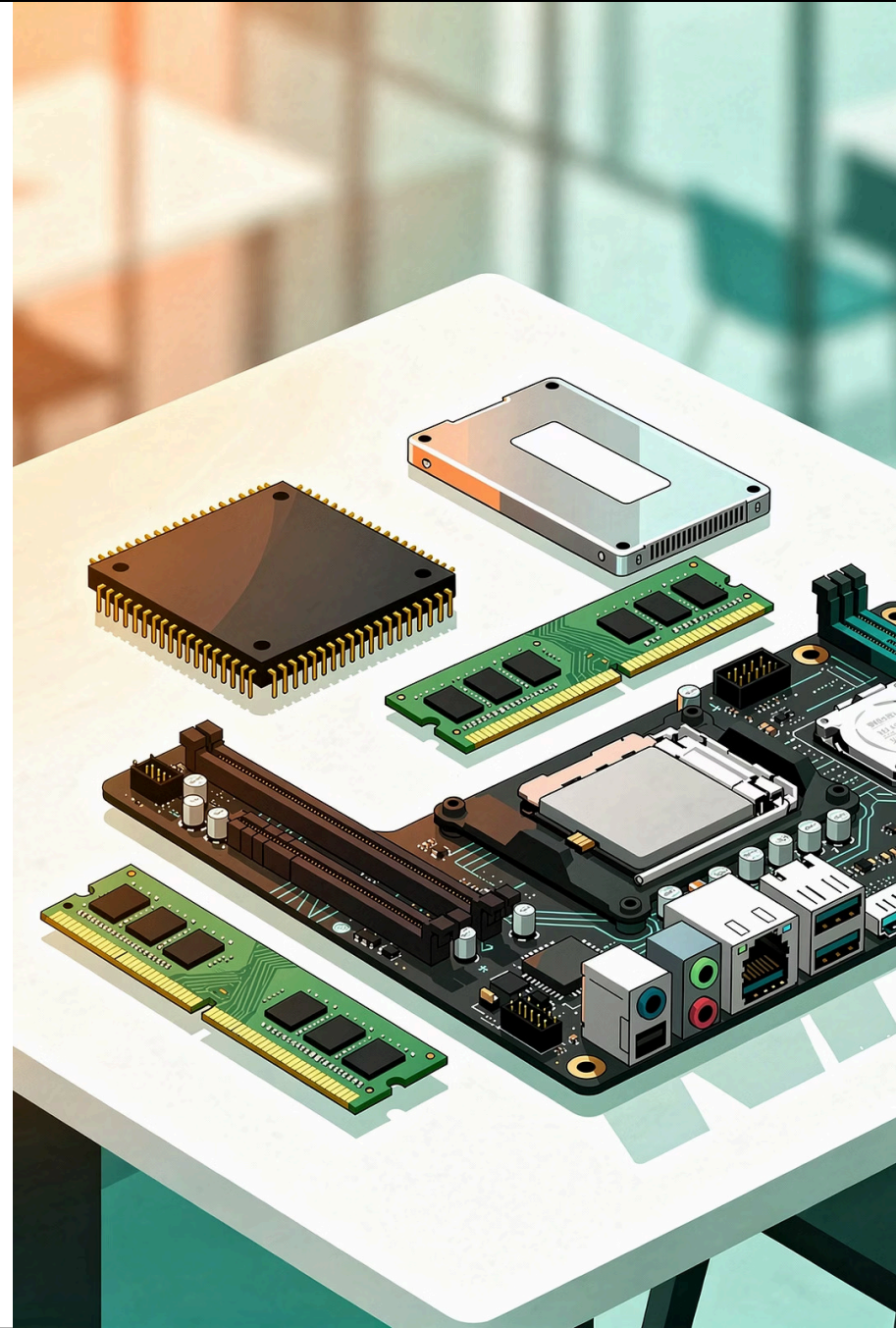
Virtual machines can be easily copied, moved, backed up, and recreated.

Foundation of the Cloud

Booked cloud servers are usually virtual machines running on real hardware in a data center.

Physical vs. Virtual Hardware

Physical hardware is everything that exists as a tangible component: processor, RAM, SSD, network card. **Virtual hardware** is an abstraction of these components provided by software—managed by the hypervisor and mapped to actual physical hardware.

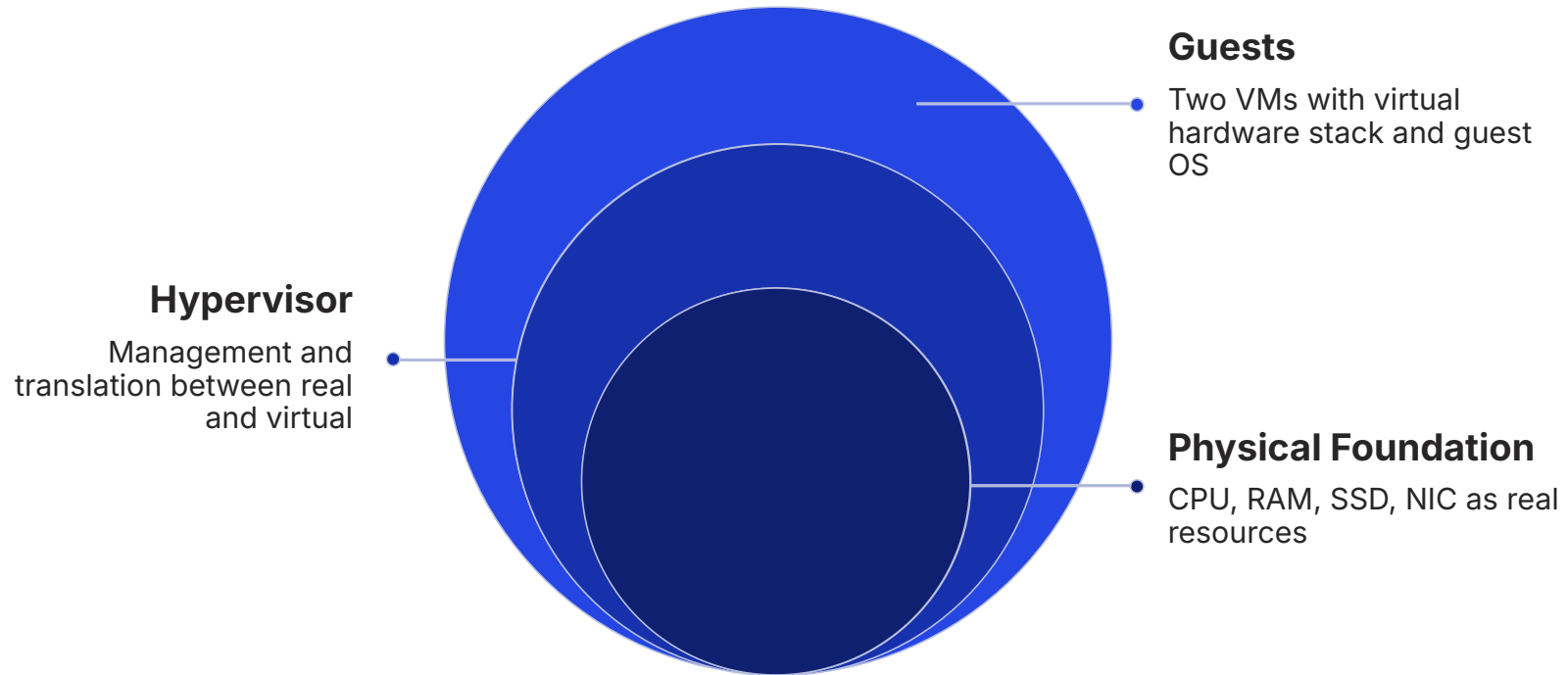


Comparison: Physical vs. Virtual

Aspect	Physical Hardware	Virtual Hardware
Existence	Actually exists as a component	Provided via software
Access	Operating system accesses directly	Access is managed via hypervisor
Quantity	Usually exists exactly once	Can be divided into many instances
Flexibility	Modifications often manual	Configuration via click or file
Portability	Device must be physically moved	VM can be moved as a file

- 
 A physical server with 32 CPU cores and 128 GB RAM can, for example, run eight VMs — each with its own vCPU and RAM allocation, without any VM "seeing" the entire server.

Layered Model: From Real to Virtual Hardware



Virtual hardware is not imaginary hardware, but a **controlled, software-defined view of actual resources**. The hypervisor is the translation and management layer between physical and virtual worlds.

Typical Use Cases

Virtualization is encountered in many situations—from large data centers to learning labs on your own laptop.



Use Cases Overview



Server Consolidation

Multiple previously separate servers are consolidated as VMs onto fewer physical hosts—saving hardware, power, and space.



Test & Development

Quickly create new VMs, try out software, and test risky changes without endangering production systems.



Cloud Infrastructure

Many cloud servers are virtual machines. Provision ten new servers in minutes—by starting new VMs.



IT Security

Isolated sandbox environments for the secure analysis of malware or risky configurations.

Additional Use Cases

Legacy Systems

Older software that only runs on specific operating system versions can continue to be operated as a VM — without needing the old physical PC.

Desktop Virtualization

Users work on a virtual desktop in the data center instead of on a local PC — simplifies administration and standardization.

Training & Learning

Run multiple operating systems on one computer, simulate networks, and make mistakes — without damaging the host system. Ideal for beginners.



Advantages for Users

Flexibility

Run multiple operating systems in parallel on the same computer — e.g., Windows as the host system and Linux in a VM.

Isolation

Errors, misconfigurations, or malware within the VM remain in their own environment — the host system remains protected.

Easy Backup

VMs are easier to back up, copy, and clone than physical systems because they are software-based.

Benefits for Enterprises

1

Better hardware utilization

Less server idle time by consolidating multiple workloads.

2

Lower costs

Fewer devices, less power, less space, and cooling required in the data center.

3

Faster provisioning

New VMs can be created in minutes — no hardware lead times.

4

Higher agility

Systems can be adapted to new requirements, migrated, and integrated into backup concepts more quickly.

Avoid a Common Misconception

Virtualization is **not a free magic trick**. All VMs share actual resources. If the host is overloaded, all VMs suffer along with it.

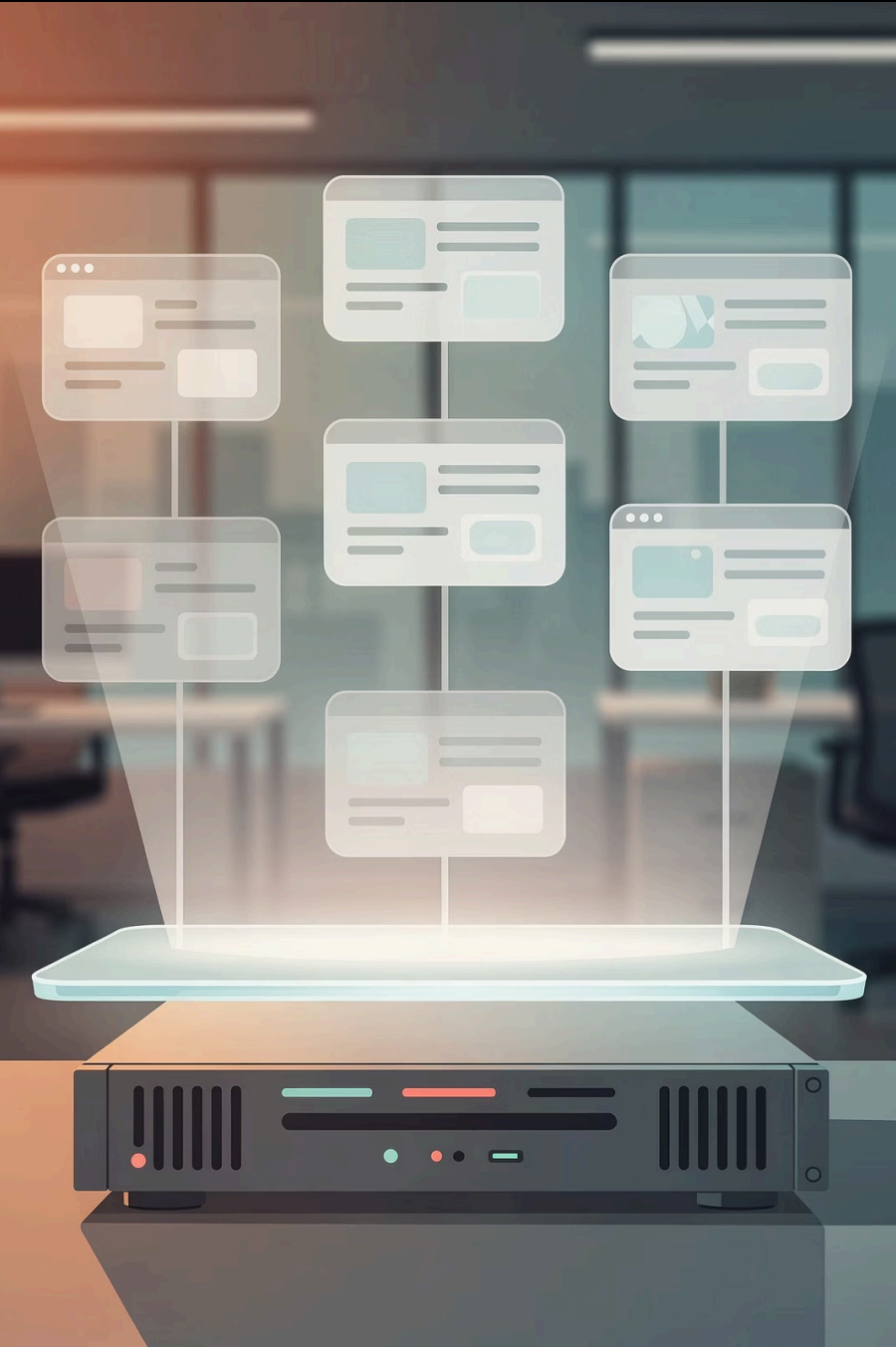
Virtualization increases efficiency — but it does not create additional physical performance out of thin air. Scalability means: You can assign more RAM or CPUs to a VM, as long as the host has enough reserves. If it is at capacity, assigning more virtual resources will not help.



CHAPTER 2

Hypervisors and Virtual Hardware

If virtualization is the core concept, then the **hypervisor** is the key technical layer that holds it all together. It creates and manages virtual machines and ensures that they can use CPU, RAM, hard drives, and networks—without uncontrolled access to physical hardware.



What a Hypervisor Does

→ **Provisioning Virtual Hardware**

The hypervisor provides each VM with virtual components such as vCPU, vRAM, vDisk, and vNIC.

→ **VM Isolation**

VMs do not directly interfere with each other — the hypervisor ensures isolation.

→ **Resource Distribution**

Physical resources are distributed and managed across various VMs.

→ **Lifecycle Management**

Startup, shutdown, configuration, snapshots, and migrations are controlled by the hypervisor.

Type 1 vs. Type 2 Hypervisor

Type 1 — Bare-Metal

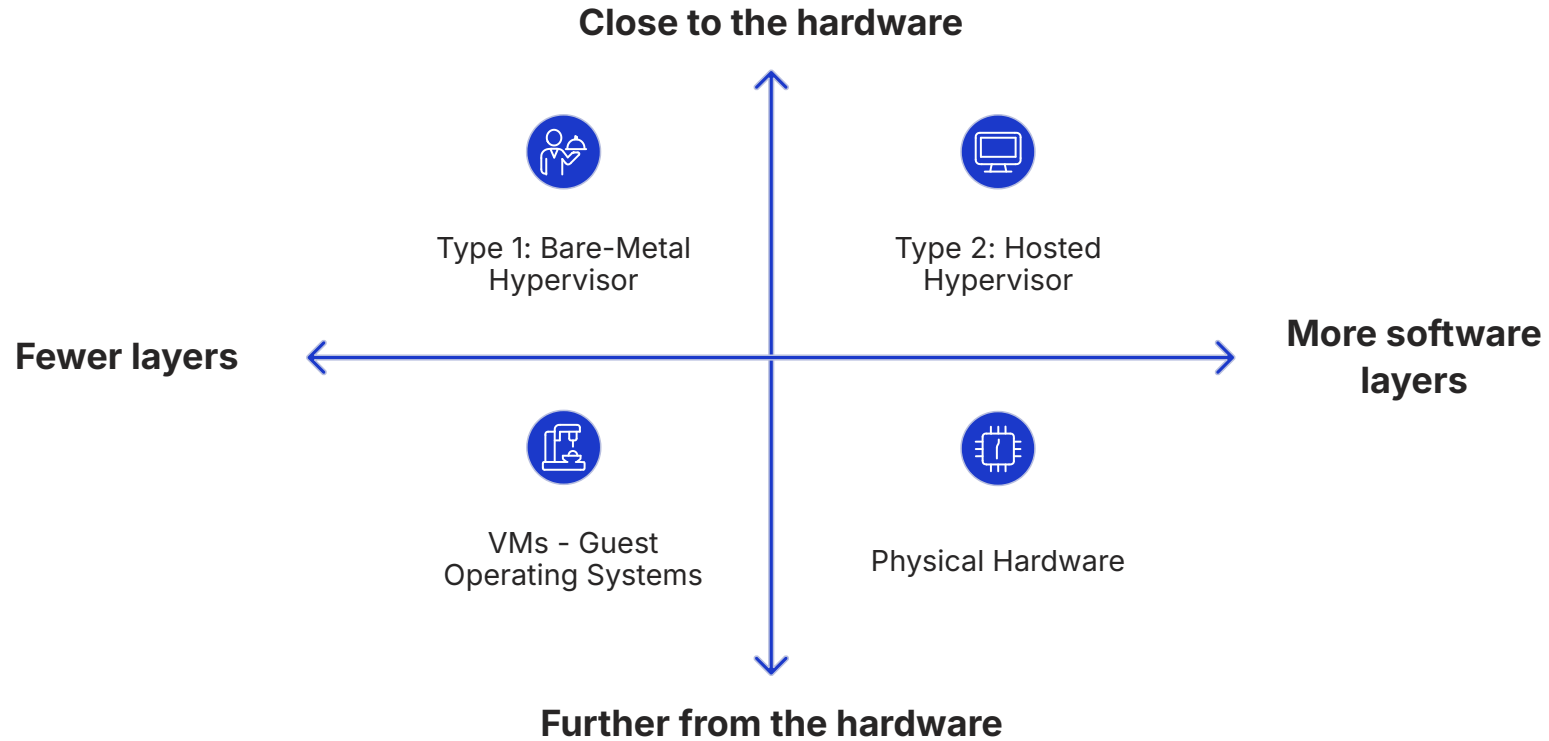
Runs **directly on the physical hardware**. Chain: Hardware → Hypervisor → VMs. Typical for data centers, server farms, and cloud. Designed more for performance, stability, and professional operation.

Type 2 — Hosted

Runs **on a host operating system**. Chain: Hardware → Host OS → Hypervisor application → VMs. Ideal for desktop use, learning, development, and testing. Easy to install like a normal application.

- ❏ Type 2 is not "worse" — it is just intended for different use cases. Often the more practical choice for labs and testing. For high-availability enterprise platforms, Type 1 is the more suitable solution.

Architecture Comparison: Type 1 and Type 2



The closer the virtualization layer is to the real hardware, the more directly and controlled it can work. This influences performance, security model, and resource management.



Rule of Thumb for Beginners

Starting VMs on your own computer?

→ Type 2 hypervisor. Just install it like a normal application and get started.

Understanding enterprise virtualization?

→ Type 1 hypervisor. Centralized management of many servers on a professional platform.

Examples of Virtualization Software

Software	Type	Common Use
VMware ESXi	Type 1	Data center, server operation
Microsoft Hyper-V	Type 1-oriented	Windows server, enterprise
KVM	Kernel-based	Linux server, cloud, hosting
Oracle VirtualBox	Type 2	Learning, development, tests
VMware Workstation	Type 2	Desktop virtualization
Parallels Desktop	Type 2	macOS desktop with guest OS

For beginners, **VirtualBox** is often the easiest choice. Those who want to understand the enterprise environment should look at **Hyper-V**, **ESXi**, or **KVM**.

VIRTUAL HARDWARE

The four virtual components of a VM

Virtual machines only appear to be real computers because they are assigned **virtual hardware components**. The guest operating system sees these components as if they were real devices — the hypervisor connects them to physical resources in the background.

Virtual CPU (vCPU)

What is a vCPU?

The vCPU is the virtual form of a processor. With 2 vCPUs, the guest operating system sees two logical processors. A vCPU is **not automatically a permanently reserved physical core** — the hypervisor schedules and distributes computing time across real processors.

Common Misconception

"4 vCPUs = 4 physical cores dedicated solely to this VM" — this does not have to be the case. If many VMs require CPU power simultaneously, they compete for the actual processor performance of the host.

- ❑ More vCPUs allow for more parallel computing work — however, too many can be unnecessary if the application does not utilize them.

Virtual RAM (vRAM)

VM Perspective

The guest operating system sees the allocated RAM (e.g., 8 GB) as if it were the computer's installed main memory.

Critical Bottleneck

RAM is often the decisive bottleneck. CPU time can be shared over time — RAM must actually be physically available.

Finding Balance

Too little RAM → guest operating system slows down. Too much RAM → is missing for other VMs or the host itself.

Virtual Disk (vDisk)

The virtual disk is to the VM what a real SSD is to a physical computer. In reality, it is often just a **file on the host system** or a storage area provided by the hypervisor.

Format	Common Environment
VDI	VirtualBox
VMDK	VMware
VHD / VHDX	Microsoft Hyper-V
qcow2	KVM/QEMU

❏ The virtual layer can abstract much — but the physical reality remains critical. A VM on a slow physical HDD will still be slow.

Virtual Network Interface Card (vNIC)

The vNIC is the VM's network connection — to the guest operating system, it looks like a standard network card. The hypervisor connects it to a virtual switch or specific network modes.

NAT

The VM accesses the network via the host — similar to being behind a router.

Bridged

The VM connects directly to the same network as other devices.

Host-only

The VM communicates only with the host or within an isolated virtual network.

Internal Network

VMs communicate only with each other — without external access. Ideal for isolated test networks.

Interaction of Virtual Components



It is the interaction that makes a VM a complete computer. The guest operating system boots, loads drivers, starts services, and works with files and network connections — almost just as it would on real hardware. **The VM does not receive magical resources, but rather virtual stand-ins for real hardware.**

Summary & Outlook

01

Virtualization Understood

Software-based abstraction of computing resources — one physical machine, many virtual systems.

02

Hypervisor as Key Layer

Type 1 (bare-metal) for data centers, Type 2 (hosted) for desktops and learning.

03

Mastering Virtual Hardware

vCPU, vRAM, vDisk, and vNIC — virtual representatives of real hardware, mediated by the hypervisor.

04

Next Topics

Snapshots, live migration, storage overcommitment, virtual switches, and cloud instances build upon these foundations.

